DO NOW — RECALL

# Operating-system functions

Name: _____ Class: _____ Date: _____

**1. What does LMC LDA do?**

**2. What does the CPU control unit do?**

**3. Define a process.**

**4. Binary search reduces a sorted 32-element candidate set to one. How many successive halvings is this?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Loads ACC from a mailbox.

**2. Model answer**

Coordinates and decodes execution.

**3. Model answer**

An instance of a program being executed or scheduled.

**4. Model answer**

5: 32→16→8→4→2→1.

**Vocabulary — know these terms**

process — Running/scheduled program instance

filesystem — Organised file storage

driver — Device interface software

spooling — Queued peripheral work

NEW LEARNING

# Read, annotate and explain

### Understand

The OS mediates application requests and access to hardware.

### Trace

Resource management schedules processes, allocates memory, controls files and drivers.

### Explain

Permissions and isolation help keep processes and users from accessing unauthorised data.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

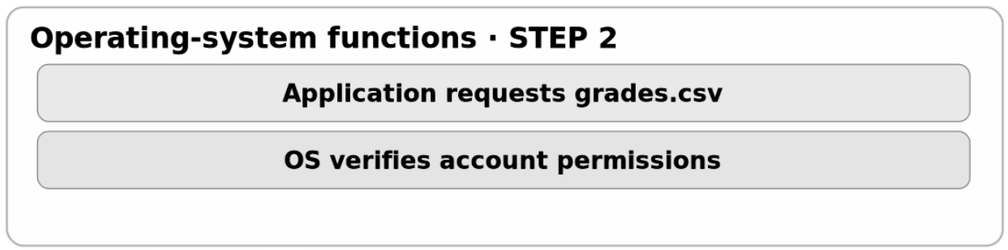
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**



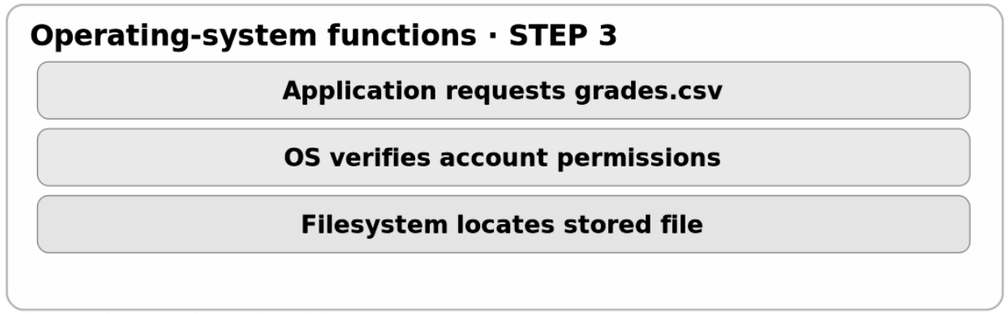

Operating-system functions · STEP 1

**New Learning**




Operating-system functions · STEP 2

**New Learning**




Operating-system functions · STEP 3





# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**An app prints a document while a printer is busy. Explain the OS's role.**

The print job enters a queue and is later sent via a compatible printer driver when the device is available.




WEDO · STATE TRACE

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

### Guided 1 — CHECK

Predict first
The print job enters a queue and is later sent via a compatible printer driver when the device is available.

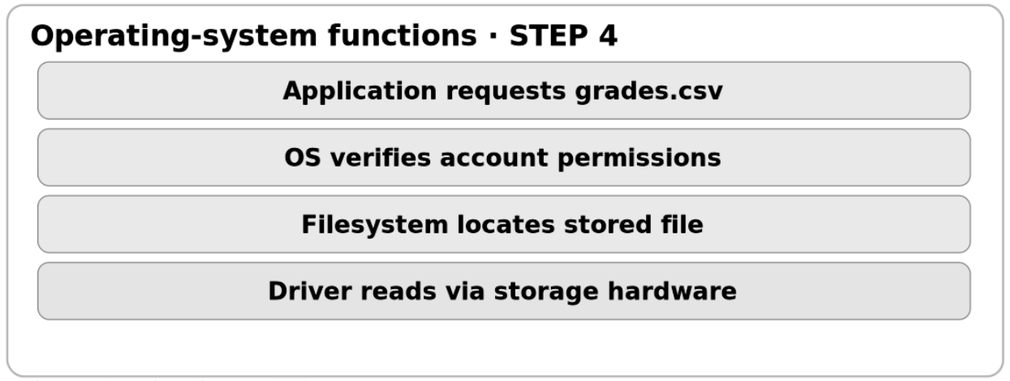




# New Learning · original lesson diagrams

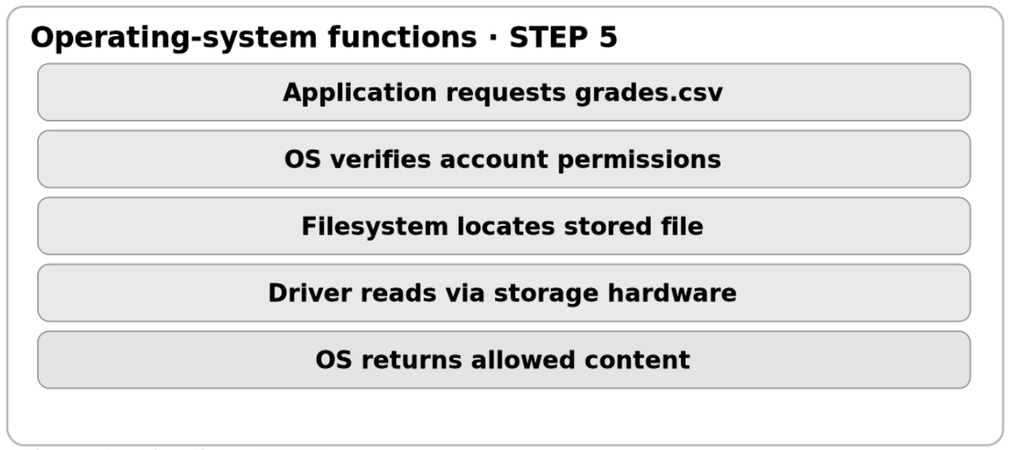
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Operating-system functions · STEP 4

**New Learning**




Operating-system functions · STEP 5

I DO — WORKED EXAMPLE

# Follow the worked method

**A pupil app requests a protected file. Explain the OS steps.**

**Predict first**

The OS checks authorisation before returning content; a missing permission should prevent the read.





EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**Applications must directly control all hardware to read files.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

**GREEN PEN — Correct and explain**

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

Explain how an OS manages a browser, game and backup concurrently, including protection. [6]





# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**A pupil app requests a protected file. Explain the OS steps.**

The OS checks authorisation before returning content; a missing permission should prevent the read.




IDO · STATE TRACE

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

An app prints a document while a printer is busy. Explain the OS's role.
Source code / input:
Application → print spool → driver → printer





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

State four operating-system responsibilities. [4]

### Question 2

Describe a protected file read from application request to result. [6]